C2 Report

Algorithm / Data Structure Explanation

I store the batches in a generic linked list because the number of batches is now known in advance. Each Batch owns a dynamically allocated array of Post objects. A deep copy is made when a batch is stored so that the analyser doesn't depend on the input array after the input array is deleted.

When a batch is added, generic merge sort is used to sort its post in ascending order of likes. This cost O(nlogn). The sorted order allows binary search to locate a post by likes and a modified binary search to find the first post hose likes are strictly greater than the threshold.

To remove a post, each batch is searched using binary search until the matching unique likes value is found. The batch then creates a smaller array and copies all posts except the removed one. Therefore, the current implementation has worst-case complexity O(b log n + n).

For a threshold query, modified binary search finds the smallest post above the threshold in each batch. The smallest of these candidates is returned, giving O(b log n)

Correctness Argument

Each batch is stored in ascending order of likes, so the modified binary search can correctly find the first post whose likes are strictly greater than the given threshold. Because the batch is sorted, this first valid post must also be the least popular post among batches. The algorithm performs this search for every batch. Whenever a valid candidate is found, it compares the candidate's number of likes with the current result. The result is updated only if the new candidate has fewer likes. Therefore, after all batches have been checked, the result points to the post with the smallest number of likes among all posts whose likes are greater than the threshold. If a batch has no post above the threshold, the modified binary search returns '-1' which indicates there is no candidate in the current batch. If no batch contains a valid post, the result remains 'nullptr'.

Testing Evidence

```
/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/C2/C2task
Enter the batch size:
3
1. Add a batch
2. Remove a post
3. Find least popular post over a threshold
4. See the menu again
5. Quit
Enter command (1-5):
1
Enter likes for post 0:
5
Enter post name for post 0:
Post_A
Enter likes for post 1:
2
Enter post name for post 1:
Post_B
Enter likes for post 2:
3
Enter post name for post 2:
Post_C
Enter command (1-5):
1
Enter likes for post 0:
10
Enter post name for post 0:
Post_D
Enter likes for post 1:
11
Enter post name for post 1:
Pose_E
Enter likes for post 2:
12
Enter post name for post 2:
Post_F
Enter command (1-5):
2
```

```
Enter likes of the post to remove:
11
Enter command (1-5):
3
Enter threshold:
10
Post found with 12 likes
Enter command (1-5):
5

Process finished with exit code 0
```